

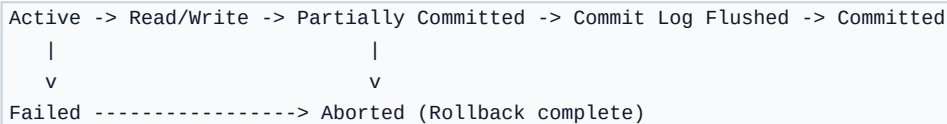
3-Mark Questions & Answers -- DBMS Module 5: Query Optimization & Transaction Concurrency Control

Q1. Explain equivalence rules for relational algebra transformations.

1. Selection Commutativity: $\sigma_{C1}(\sigma_{C2}(E)) \equiv \sigma_{C2}(\sigma_{C1}(E))$

2. Jo

Q2. Explain transaction state transition diagram.



Q3. Differentiate between Strict 2PL and Rigorous 2PL.

Feature	Strict 2PL	Rigorous 2PL
Exclusive Locks (X)	Held until Transaction Commit/Abort.	Held until Transaction Commit/Abort.
Shared Locks (S)	Released during Shrinking phase.	Held until Transaction Commit/Abort.
Cascading Rollback	Prevents cascading aborts.	Prevents cascading aborts + strict ser

Q4. Explain the Validation-Based (Optimistic) concurrency control protocol.

Optimistic Protocol assumes conflicts are rare and executes in 3 phases:

1. Re

Q5. Explain Write-Ahead Logging (WAL) protocol.

The Write-Ahead Logging (WAL) protocol dictates that log records describing data modifications MUST be written and flushed to non-volatile disk storage BEFORE the actual dirty database memory blocks are written to disk.